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General Comments : Consolidated HW02 submission document uniting the UP Inception Overview (Vision, Personas, Requirements, Business Rules, Use Cases, Glossary) and the Agile Scrum Product Backlog, Themes/Epics, User Stories, Tasks, Sprint Plan, and PSI/MVP definition for the CSE305 Library Management System.

HW02 — UP Inception Overview & Agile Scrum Product Backlog

Library Management System (LMS)

Course	CSE305 Software Engineering — Adnan Menderes University (ADU), Department of Computer Engineering
Instructor	Assoc. Prof. Dr. Fatih Soygazi
Project	Library Management System (LMS)
Team & Scrum roles	**Ibrahim Temur** (`231805033`) — Product Owner · **Omer Faruk Kuş** (`231805026`) — Scrum Master (both: Developer, Architect, QA, Reviewer)
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Archive	`cse305_hw02_231805033_231805026.zip` (created with `python3 tools/package.py 02` / `tools\package 02`)
GitHub repository	`https://github.com/ibrahimtemur/cse305-library-management-system`

> **Note**
> **How to read this document.** Part 1 provides a self-contained executive consolidation of the Unified Process (UP) Inception artifacts and links to the authoritative detail files in `docs/02-inception/` : `docs/02-inception/VISION_AND_CONTEXT.md` , `docs/02-inception/PERSONAS.md` , `docs/02-inception/REQUIREMENTS.md` (`FR-001` .. `FR-032` , `NFR-001` .. `NFR-012` , `CON-001` .. `CON-006` , `DR-001` .. `DR-007`), `docs/02-inception/BUSINESS_RULES.md` (`BR-001` .. `BR-018`), `docs/02-inception/USE_CASES.md` (`UC-01` .. `UC-11`), and `docs/02-inception/DOMAIN_GLOSSARY.md` . Part 2 presents the complete Agile Scrum deliverables: Themes & Epics (`docs/02-inception/THEMES_EPICS.md`), User Stories (`docs/02-inception/USER_STORIES.md` , `LMS-001` .. `LMS-021`), Ordered Product Backlog (`docs/02-inception/PRODUCT_BACKLOG.md`), Three-Sprint Iteration Plan (`docs/02-inception/SPRINT_PLAN.md`), Task Breakdown (`docs/02-inception/TASKS.md` , `105` tasks), and Potentially Shippable Increment / MVP Definition (`docs/02-inception/PSI_AND_MVP.md`). Every story point, classic hour, human hour, capacity calculation, commitment line, and traceability check is verified deterministically by `tools/agile/validate_backlog.py` .

Contents: Part 1 — UP Inception Overview · Part 2 — Agile Scrum Product Backlog & Iteration Plan · Revision History

Part 1 — UP Inception Overview

1.1 Vision, Problem Recap, Goals & System Context

Full specification: `docs/02-inception/VISION_AND_CONTEXT.md` .

Product Vision Statement

For university, faculty, and departmental libraries that manage physical book collections for academic staff, administrative staff, and students, **who** struggle with error-prone manual ledgers, untracked physical book copies, inconsistent overdue penalty calculations, and lack of real-time shelf visibility, **the Library Management System (LMS)** is a self-contained, web-based library catalog and circulation management application **that** automates bibliographic cataloging, barcode-level physical copy lifecycle tracking (`AVAILABLE` , `LOANED` , `RESERVED` , `MAINTENANCE` , `LOST`), policy-driven lending and renewal rules, First-In-First-Out (FIFO) title reservation queues, and deterministic overdue fine calculation. **Unlike** informal spreadsheet tracking or heavyweight commercial integrated library systems requiring external institutional infrastructure, **our product** operates 100% offline out-of-the-box with an embedded H2 database seeded from public-domain Open Library metadata (`782` books, `1,706` physical copies) and strictly synthetic operational data (`60` members, `3` librarians, `164` loans, `47` fines, `22` reservations), enforcing the confirmed Adnan Menderes University (ADU) library lending rules through modular GoF/GRASP design patterns and a JUnit 5 Test-Driven Development (TDD) test suite.

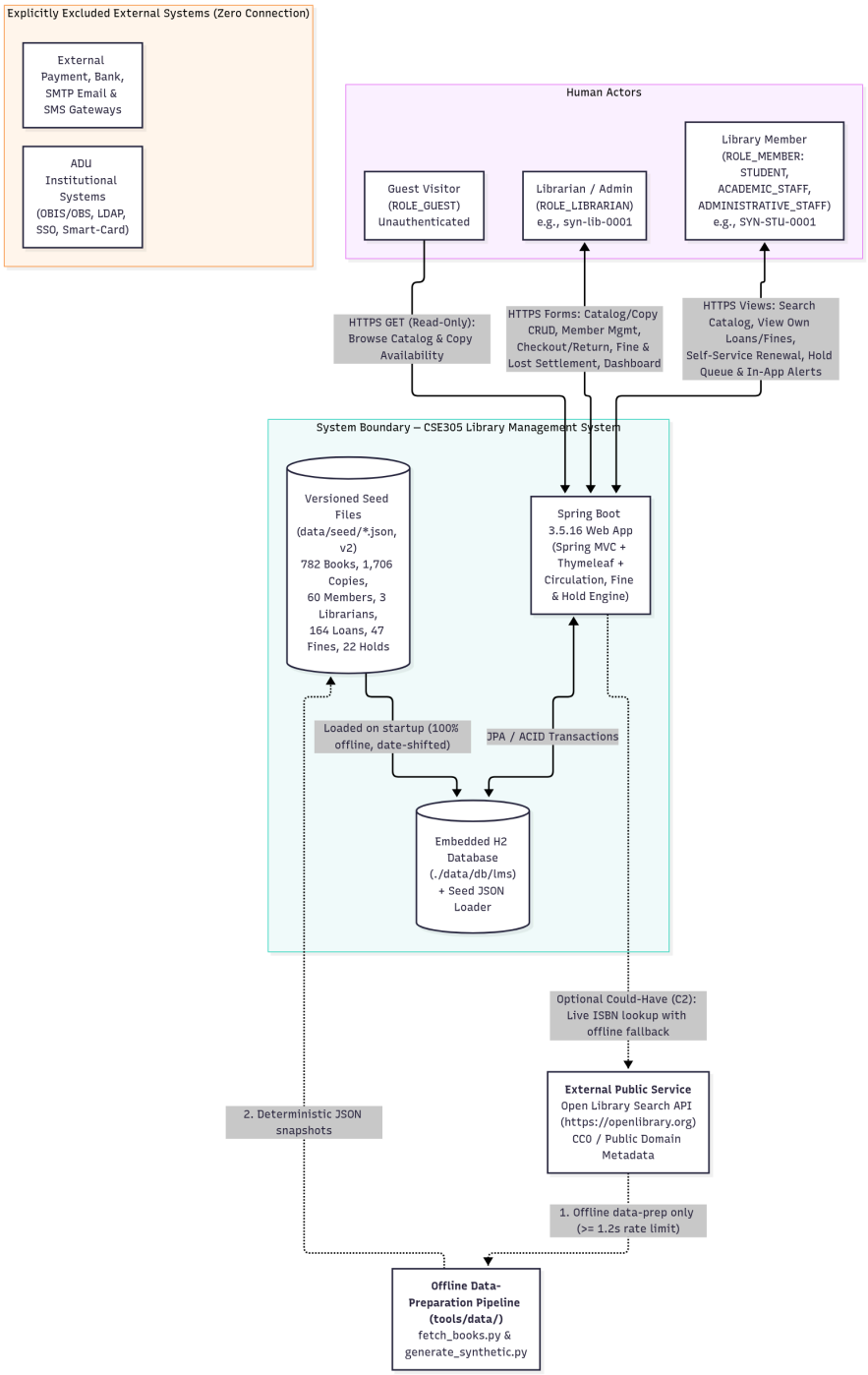
Problem Recap & Project Goals (`GOAL-01` .. `GOAL-05`)

Small-to-medium academic libraries using paper logs or unvalidated spreadsheets face four bottlenecks: (1) clerical overhead and manual date/ID errors at checkout/return, (2) inability to distinguish shelf-available copies from loaned, reserved, maintenance, or lost copies (or non-circulating reference works), (3) inconsistent enforcement of member quotas (`7` / `3` / `3`), `0.25` `TRY/day` fines, post-renewal cooldowns, and borrowing blocks, and (4) zero self-service visibility for members.

Goal / Metric ID	Category	Summary Statement & Verification Basis
`GOAL-01` / `MET-01`	**Operational Automation & Policy Accuracy**	Replace manual ledgers with transactional checkout, return, renewal, reservation, and fine settlement governed by `BR-001` .. `BR-018` . **Target (`ASSUMPTION`):** `100%` pass rate across JUnit 5 tests covering quotas (`7` / `3` / `3`), `30` -day periods, `0.25 TRY/day` uncapped fines, `30` -day post-renewal cooldown, and blocks.
`GOAL-02` / `MET-02`	**Copy-Level Inventory & State Integrity**	Track every physical `BookCopy` (`LMS-COPY-1001` ..) via `BookCopyState` and block checkout of `referenceOnly` books (`BR-014` , `BR-017`). **Target (`ASSUMPTION`):** `0` illegal transitions or double-loaned copies; `100%` 1-to-1 invariant between `LOANED` copies and `ACTIVE` loans, and between `RESERVED` copies and `READY` holds.
`GOAL-03` / `MET-03`	**Self-Service & Local Query Latency**	Empower members to search the catalog, inspect loans/fines/cooldowns, renew, and place/cancel holds (`UC-01` , `UC-07` , `UC-08` , `UC-10`). **Target (`ASSUMPTION`):** `< 200 ms` local server response time on the reference workstation against the seeded H2 dataset (`NFR-005`).
`GOAL-04` / `MET-04`	**CSE305 Engineering Rigor & Startup Speed**	Apply UP modeling, Agile Scrum, UI-First prototyping via **Stitch MCP** (`NFR-001`), GoF/GRASP patterns (`NFR-010`), and TDD (`NFR-011`). **Target (`ASSUMPTION`):** `< 10 seconds` offline startup and seed initialization time with zero runtime internet calls (`NFR-004` , `NFR-006`).
`GOAL-05` / `MET-05`	**Offline Reproducibility & Synthetic Privacy**	`0` real personal names, student IDs, emails, or plaintext secrets in Git history; `100%` of seeded emails use `@synthetic.lms.local` and demo passwords are BCrypt-hashed at startup (`NFR-008` , `NFR-009` , `BR-018`).

System Scope & Explicit Exclusions

- **In Scope:** Printed books only (including non-circulating `referenceOnly` encyclopedias, dictionaries, and catalogs, `BR-014`); copy-level inventory (`1,706` seeded copies); synthetic `STUDENT` , `ACADEMIC_STAFF` , and `ADMINISTRATIVE_STAFF` members; desk checkout/return, self-service/desk renewal, `0.25 TRY/day` uncapped fine calculation, desk payment/waiver, lost-copy petition (`lostReportedOn` , fine freeze, 30-day replacement window, `2x` runtime `appraisedValue`), FIFO title reservation queue with 3-day pickup hold and in-app notification, librarian KPI dashboard, and Stretch enhancements (`ROLE_GUEST` portal, CSV export, live Open Library ISBN lookup helper).
- **Explicitly Out of Scope (`CON-002` .. `CON-004`):** (1) **Periodicals/serials and unpublished theses** (although ADU regulations allow borrowing 2 non-latest periodical issues for 2 days and restrict unpublished theses, LMS manages books only); (2) **Live ADU OBIS/OBS, LDAP/SSO, or RFID/smart-card hardware** (personal ID card verification `BR-016` is performed by the Librarian at the desk using `memberId`); (3) **External payment/banking APIs and external SMTP email/SMS gateways** (fines are settled at the desk as `PAID` / `WAIVED` ; hold-ready alerts are rendered in-app).



1.2 Stakeholders & User Personas

Full specification and circulation scenarios: `docs/02-inception/PERSONAS.md`. Every persona is anchored to real synthetic records in `data/seed/*.json`.

#	Persona & Role	Member Type (`memberType`)	Representative Seed Record (`data/seed/*.json`)	Confirmed Quota (BR-002)	Confirmed Period (BR-003)	Core Operational / Self-Service Goals
1	**Librarian / Admin** (`ROLE_LIBRARIAN`)	Staff Account (`ASSUMPTION-01`)	`syn-lib-0001` (`Enginiz Yıldırım`) in `librarians.json` *(colleagues: `syn-lib-0002`, `syn-lib-0003`)*	N/A	N/A	Fast desk checkout/return by `memberId` & `barcode`, catalog/copy CRUD, member management, fine settlement/waiver (`settledBy`), lost-copy assessment, KPI dashboard (`UC-03`..`UC-07`, `UC-09`, `UC-11`).

#	Persona & Role	Member Type (`memberType`)	Representative Seed Record (`data/seed/*.json`)	Confirmed Quota (`BR-002`)	Confirmed Period (`BR-003`)	Core Operational / Self-Service Goals
2	**Student Member** (`ROLE_MEMBER`)	`STUDENT`	`SYN-STU-0001` (`Tutkucan Ergül`, Chemistry, `LOAN-0099`) & `SYN-STU-0003` (`Hazrat Yorulmaz`, Computer Eng., `RES-0009` `READY` on `LMS-COPY-1599`)	**3 books**	**30 days**	Search course textbooks, check availability & `referenceOnly` badges, view own active/historical loans, cooldowns & fines, renew once online, join FIFO hold queue (`UC-01`, `UC-07`, `UC-08`, `UC-10`).
3	**Academic Staff Member** (`ROLE_MEMBER`)	`ACADEMIC_STAFF`	`SYN-STF-0003` (`Atasagun Öcalan`, Computer Eng., `RES-0007`) & `SYN-STF-0002` (`Baydu Arsoy`, Mathematics, `RES-0001`)	**7 books**	**30 days**	Borrow up to `7` research titles simultaneously (`AcademicStaffLoanPolicy`), inspect copy counts for course reading lists, renew non-reserved titles, reserve high-demand literature (`UC-01`, `UC-07`, `UC-08`, `UC-10`).
4	**Administrative Staff Member** (`ROLE_MEMBER`)	`ADMINISTRATIVE_STAFF`	`SYN-ADM-0001` (`Doğannur Güçlü`, Business Admin.) & `SYN-ADM-0004` (`Peren Erdoğan`, Computer Eng.)	**3 books**	**30 days**	Browse general & professional books, borrow up to `3` books in person with university staff ID card (`AdministrativeStaffLoanPolicy`), track due dates and renew online (`UC-01`, `UC-07`, `UC-10`).
5	**Guest Visitor** (`ROLE_GUEST`)	Unauthenticated	Anonymous browser session	**0 books**	N/A	Browse and filter the public catalog and inspect shelf availability without logging in, with zero access to member or loan data (`UC-01`, `FR-030`).

1.3 Supplementary Specification Overview (`FR`, `NFR`, `CON`, `DR`)

Full specification of all `32` Functional Requirements, `12` Non-Functional Requirements, `6` Constraints, and `7` Data Requirements: `docs/02-inception/REQUIREMENTS.md` and `data/README.md`.

Summary of Requirement Categories & Seed v2 Counts

Category	ID Range	Count	Breakdown & Scope Summary
Functional Requirements	`FR-001` .. `FR-032`	**32**	**19 Must Have** (`FR-001` .. `FR-019` across `M1` – `M6`), **10 Should Have** (`FR-020` .. `FR-029` across `S1` – `S4`), **3 Could Have** (`FR-030` .. `FR-032` across `C1` – `C3`).
Non-Functional Requirements	`NFR-001` .. `NFR-012`	**12**	Usability (`NFR-001` Stitch MCP UI-first, `NFR-002` desk barcode efficiency), Reliability (`NFR-003` `@Transactional` ACID, `NFR-004` 100% offline runtime), Performance targets (`NFR-005` `< 200 ms`, `NFR-006` `< 10 s` startup), Security/Privacy (`NFR-007` RBAC/validation, `NFR-008` BCrypt/no secrets, `NFR-009` synthetic PII), Maintainability/Testability/Portability (`NFR-010` .. `NFR-012`).
System Constraints	`CON-001` .. `CON-006`	**6**	English & mandatory header policy (`CON-001`), books-only scope (`CON-002`), no live ADU integration (`CON-003`), no payment/SMTP/SMS gateways (`CON-004`), `tools/set-author` shared-PC rule (`CON-005`), protected `main` & dual ADUZEM upload (`CON-006`).
Seed Data Requirements	`DR-001` .. `DR-007`	**7**	`books-openlibrary.json` (**782 books**, 29 `referenceOnly`), `copies.json` (**1,706 copies**: 1,472 `AVAILABLE`, 164 `LOANED`, 8 `RESERVED`, 49 `MAINTENANCE`, 13 `LOST`), `members.json` (**60 members**: 42 `STUDENT`, 12 `ACADEMIC_STAFF`, 6 `ADMINISTRATIVE_STAFF`; 56 `ACTIVE`, 4 `SUSPENDED`), `librarians.json` (**3 librarians**), `loans.json` (**164 loans**: 58 `ACTIVE`, 93 `RETURNED`, 13 `LOST`), `fines.json` (**47 fines**: 18 `UNPAID`, 23 `PAID`, 6 `WAIVED`), `reservations.json` (**22 reservations**: 14 `WAITING`, 8 `READY`).

Highlights of the Most Critical Functional & Non-Functional Requirements

ID(s)	Priority / Category	Condensed Requirement Statement	Linked `BR-*` / `UC-*`
`FR-001` .. `FR-005`	Must (`M1` Catalog & Copies)	Librarian CRUD on `Book` (`isbn13` checksum, 11 categories, `referenceOnly` flag for encyclopedias/dictionaries/catalogs), public/member search & detail view with `REFERENCE ONLY` badge, barcode registration (`LMS-COPY-*`), and GoF **State** machine (`BookCopyState`) over `AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST`.	`BR-001`, `BR-014`, `BR-015`, `BR-017` · `UC-01`, `UC-03`

ID(s)	Priority / Category	Condensed Requirement Statement	Linked BR-* / UC-*
FR-009 .. FR-013	Must (M3 Checkout & Return)	Atomic `@Transactional` desk checkout verifying personal ID (`memberId` , no proxy), `ACTIVE` status, quota (7 / 3 / 3), zero overdue loans, zero `UNPAID` fines, 30-day post-renewal cooldown on `isbn13` , and copy eligibility (`!referenceOnly` , `AVAILABLE` or matching `READY` hold); atomic return computing overdue fines and setting copy state.	`BR-002` , `BR-003` , `BR-006` .. `BR-008` , `BR-011` , `BR-013` , `BR-014` , `BR-016` , `BR-017` · `UC-05` , `UC-06`
FR-014 , FR-015	Must (M4 Overdue & Fines)	Zero-grace-period overdue calculation `overdueDays = max(0, (effectiveEndDate - dueDate).days)` (using `lostReportedOn` , `returnDate` , or `currentDate`) and uncapped daily fine `amount = 0.25 * overdueDays` TRY via `FineCalculationStrategy` .	`BR-007` .. `BR-010` , `BR-015` · `UC-06` , `UC-09` , `UC-11`
FR-016 .. FR-019	Must (M5 Auth/Self-Service & M6 Seed Loader)	Local login/logout with startup-seeded BCrypt demo passwords (`OPEN-05`), endpoint RBAC (`ROLE_LIBRARIAN` , `ROLE_MEMBER` , `ROLE_GUEST`), member self-service portal + librarian inspection (`FR-008` , `FR-018`), and startup JSON seed loader with date shifting (`delta = today - 2026-10-22` , `OPEN-04`).	`BR-001` , `BR-002` , `BR-006` , `BR-010` , `BR-013` , `BR-017` , `BR-018` · `UC-02` , `UC-04` , `UC-10`
FR-020 .. FR-023	Should (S1 FIFO Hold Queue)	Title-level reservation when `0` copies are `AVAILABLE` and `!referenceOnly` ; strict FIFO `queuePosition = 1..n` ; GoF **Observer** hold promotion on return (`copy -> RESERVED` , `reservation -> READY` , `holdUntil = readyOn + 3 days` , in-app notification); cancellation & 3-day expiry reallocation.	`BR-011` .. `BR-014` · `UC-06` , `UC-08` , `UC-10`
FR-024 .. FR-029	Should (S2 Renewal, S3 Fine/Lost, S4 Dashboard)	Single `+30` -day renewal if not overdue and unreserved (`FR-024`); desk fine `PAID` and `WAIVED` settlement (`FR-025` , `FR-026`); lost-copy petition (`lostReportedOn` freezes fine, 30-day window, `2.00 * appraisedValue` , `FR-027`); dashboard KPI cards (`FR-028`) and overdue/unpaid report (`FR-029`).	`BR-003` .. `BR-005` , `BR-007` .. `BR-011` , `BR-015` , `BR-017` · `UC-07` , `UC-09` , `UC-11`
NFR-001 , 003 , 004 , 008 .. 011	Non-Functional Core	UI-first Stitch MCP prototypes in `docs/03-design/ui/` before Thymeleaf views; `@Transactional` ACID integrity; 100% offline build & test execution; zero committed passwords/PII; 5-tier layered architecture with GoF/GRASP patterns and JUnit 5 + Python `unittest` suites.	Architecture & QA gates

1.4 Business Rules Overview (`BR-001` .. `BR-018`)

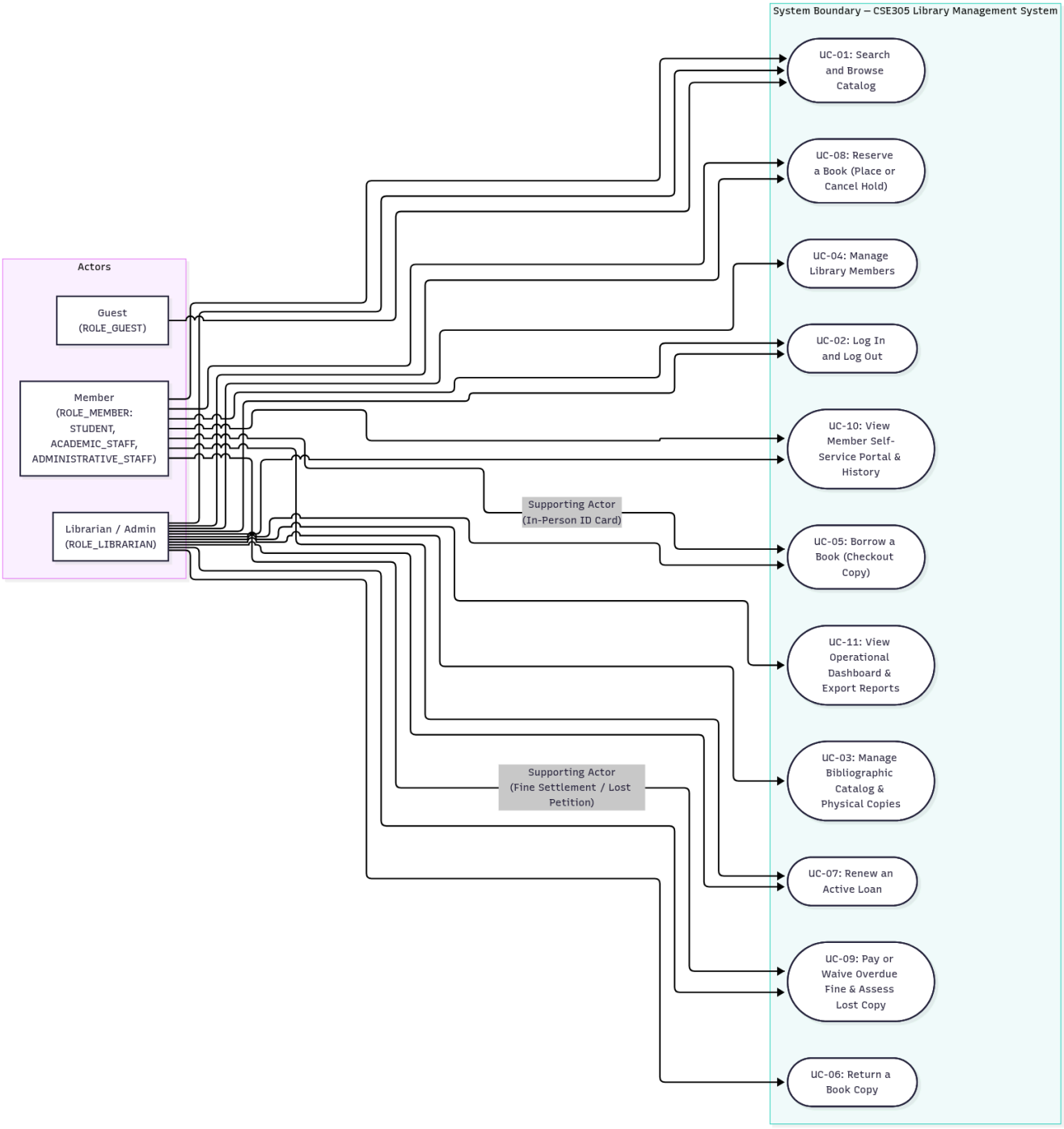
Full catalog of all `18` business rules, official ADU source citations (verified `2026-10-22`), HW01 assumption disposition table (`ASSUMPTION-01` .. `09`), new assumptions (`ASSUMPTION-10` .. `17`), and exact `BigDecimal` / `LocalDate` formulas: `docs/02-inception/BUSINESS_RULES.md` .

Rule ID(s)	Rule Name	Summary of Rule & Deterministic Formula	Verification Status
BR-002 , BR-003	**Borrowing Quotas & 30-Day Loan Period**	Max simultaneous `ACTIVE` loans: **`ACADEMIC_STAFF = 7`** , **`ADMINISTRATIVE_STAFF = 3`** , **`STUDENT = 3`** . Standard loan duration is **`30 calendar days`** for all three member types (`dueDate = checkoutDate + 30 days`).	**`CONFIRMED`** *(ADU Official Rules, 2026-10-22)*
BR-004 , BR-005	**Single Renewal & Renewal Prohibitions**	An `ACTIVE` loan may be extended at most **once** (`renewalCount: 0 -> 1` , `dueDate = checkoutDate + 60 days`), and **cannot** be renewed if `currentDate > dueDate` (overdue) or if any `WAITING` / `READY` reservation exists on `isbn13` .	**`CONFIRMED`** (`BR-004`) + `ASSUMPTION-11` (`BR-005` : max `1` renewal of `+30` days)
BR-006	**Post-Renewal 1-Month Re-Borrowing Cooldown**	When a renewed loan (`renewalCount >= 1`) is returned on `returnDate` , the same `memberId` cannot borrow any copy of the same `isbn13` until `returnDate + 30 calendar days` (`currentDate < returnDate + 30 days` is rejected).	**`CONFIRMED`** *(1-month rule)* + `ASSUMPTION-12` *(30-day mapping)*
BR-007 .. BR-010	**Overdue Days, `0.25 TRY/day` Tariff (No Cap) & Settlement**	Zero grace period: `overdueDays = max(0, (effectiveEndDate - dueDate).days)` . Daily fine is **`0.25 TRY` (25 kuruş) per overdue day** with **no maximum cap** (`amount = 0.25 * overdueDays`). At most one `Fine` per `loanId` ; settled as `PAID` or `WAIVED` by a Librarian (`settledOn` , `settledBy`) after loan resolution.	**`CONFIRMED`** (`0.25 TRY/day`) + `ASSUMPTION-10` *(no cap; `250 TRY` cap removed)*
BR-011	**Borrowing & Renewal Block**	Any member holding **`>= 1 overdue active loan`** OR **`>= 1 UNPAID fine ( amount > 0.00 TRY )`** OR `status == SUSPENDED` is blocked from borrowing new books or renewing existing loans.	**`CONFIRMED`** *(overdue & unpaid fine block)* + `ASSUMPTION-06` *(SUSPENDED)*
BR-012 , BR-013	**FIFO Reservation Queue & 3-Day Pickup Hold**	Title-level hold (`isbn13`) allowed only when `0` copies are `AVAILABLE` (and member is `ACTIVE` , not already borrowing/queuing for `isbn13`). On copy return, `copy -> RESERVED` , `queuePosition == 1` reservation -> `READY` (`readyCopyBarcode` , `readyOn = returnDate` , `holdUntil = readyOn + 3 days`), and an **in-app notification** is created.	**`CONFIRMED`** *(notification & 2–3 day hold)* + `ASSUMPTION-13` *(FIFO, in-app, 3 days)*

Rule ID(s)	Rule Name	Summary of Rule & Deterministic Formula	Verification Status
BR-014	**Non-Circulating Reference Works (referenceOnly)**	**Encyclopedias, dictionaries, and catalogs** have referenceOnly = true and cannot be borrowed or reserved. *(Periodicals and unpublished theses are Out of Scope per C0N-002).*	**CONFIRMED** *(ADU Official Rules)* + ASSUMPTION-14 *(referenceOnly boolean)*
BR-015	**Lost-Copy Petition, Fine Freeze & 2x Valuation Penalty**	Filing a lost-book petition (lostReportedOn) **immediately freezes daily overdue fine accrual on lostReportedOn** and opens a 30-day replacement window. If unreplaced after 30 days, copy -> LOST , loan -> LOST , and the Librarian enters the Valuation Commission's appraisedValue at runtime to assess **compensationAmount = 2.00 * appraisedValue** plus frozen overdue fine.	**CONFIRMED** *(freeze, 1-month window, 2x valuation)* + ASSUMPTION-12 , ASSUMPTION-15
BR-016 .. BR-018	**Personal ID Card, Copy State Invariant & Synthetic Auth**	Borrowers must present their own personal university ID card (memberId ; **no proxy borrowing**, BR-016). Every copy has a single valid state (BR-017). Seed JSON contains 100% synthetic PII and zero passwords; BCrypt demo passwords are initialized at startup (BR-018).	**CONFIRMED** (BR-016) + ASSUMPTION-08 , 09 , 16 , 17

1.5 Use Case Model Overview (UC-01 .. UC-11)

Full specifications (including the **4 fully-dressed** use cases UC-01 , UC-05 , UC-06 , UC-09 with preconditions, postconditions, main success scenarios, and alternative/exception flows, plus the **7 brief** use cases): docs/02-inception/USE_CASES.md .



UC ID	Use Case Name	Primary Actor(s)	Detail Level in `USE_CASES.md`	Realized `FR-*` IDs	Enforced `BR-*` IDs	Implementing Stories
`UC-01`	**Search and Browse Catalog**	`Guest`, `Member`, `Librarian`	**Fully-Dressed**	`FR-002`, `FR-003`, `FR-030`	`BR-001`, `BR-014`, `BR-017`	`LMS-004` (S1), `LMS-018` (S3)
`UC-02`	**Log In and Log Out**	`Member`, `Librarian`	Brief	`FR-016`, `FR-017`	`BR-001`, `BR-018`	`LMS-003` (S1)
`UC-03`	**Manage Bibliographic Catalog & Physical Copies**	`Librarian`	Brief	`FR-001`, `FR-004`, `FR-005`, `FR-031`	`BR-001`, `BR-014`, `BR-017`	`LMS-005` (S1), `LMS-020` (S3)
`UC-04`	**Manage Library Members**	`Librarian`	Brief	`FR-006`, `FR-007`, `FR-008`	`BR-001`, `BR-002`, `BR-011`, `BR-018`	`LMS-006` (S2), `LMS-013` (S2)
`UC-05`	**Borrow a Book (Checkout Copy)**	`Librarian` *(Member in person)*	**Fully-Dressed**	`FR-005`, `FR-009` .. `FR-011`, `FR-013`	`BR-002`, `BR-003`, `BR-006`, `BR-011`, `BR-013`, `BR-014`, `BR-016`, `BR-017`	`LMS-007` (S2), `LMS-017` (S3 hold pickup)

UC ID	Use Case Name	Primary Actor(s)	Detail Level in USE_CASES.md	Realized FR-* IDs	Enforced BR-* IDs	Implementing Stories
UC-06	**Return a Book Copy**	Librarian	**Fully-Dressed**	FR-005, FR-012 .. FR-015, FR-022	BR-006 .. BR-010, BR-013, BR-017	LMS-010 (S2), LMS-017 (S3 hold promo)
UC-07	**Renew an Active Loan**	Member, Librarian	Brief	FR-024	BR-003 .. BR-006, BR-011	LMS-014 (S3)
UC-08	**Reserve a Book (Place or Cancel Hold)**	Member, Librarian	Brief	FR-020 .. FR-023	BR-011 .. BR-014	LMS-017 (S3), LMS-012 (S3)
UC-09	**Pay or Waive Overdue Fine & Assess Lost Copy**	Librarian *(Member)*	**Fully-Dressed**	FR-005, FR-014, FR-015, FR-025 .. FR-027	BR-007 .. BR-011, BR-015, BR-017	LMS-009 (S2), LMS-011 (S3), LMS-016 (S3)
UC-10	**View Member Self-Service Portal & History**	Member, Librarian	Brief	FR-008, FR-018	BR-001, BR-002, BR-006, BR-010, BR-013	LMS-013 (S2), LMS-017 (S3 holds view)
UC-11	**View Operational Dashboard & Export Reports**	Librarian	Brief	FR-014, FR-015, FR-028, FR-029, FR-032	BR-001, BR-007, BR-008, BR-011	LMS-015 (S3), LMS-021 (S3), LMS-019 (S3)

1.6 Domain Glossary & Conceptual Domain Model Bridge

Full ubiquitous language glossary (20 entity/policy definitions + 8 enumeration tables): docs/02-inception/DOMAIN_GLOSSARY.md. Conceptual domain model (7 classes, 50 attributes, 6 enums, 8 associations A1 .. A8, and narrative traceability D1 .. D8): docs/01-proposal/HW01_Problem_Solution_DomainModel.md.

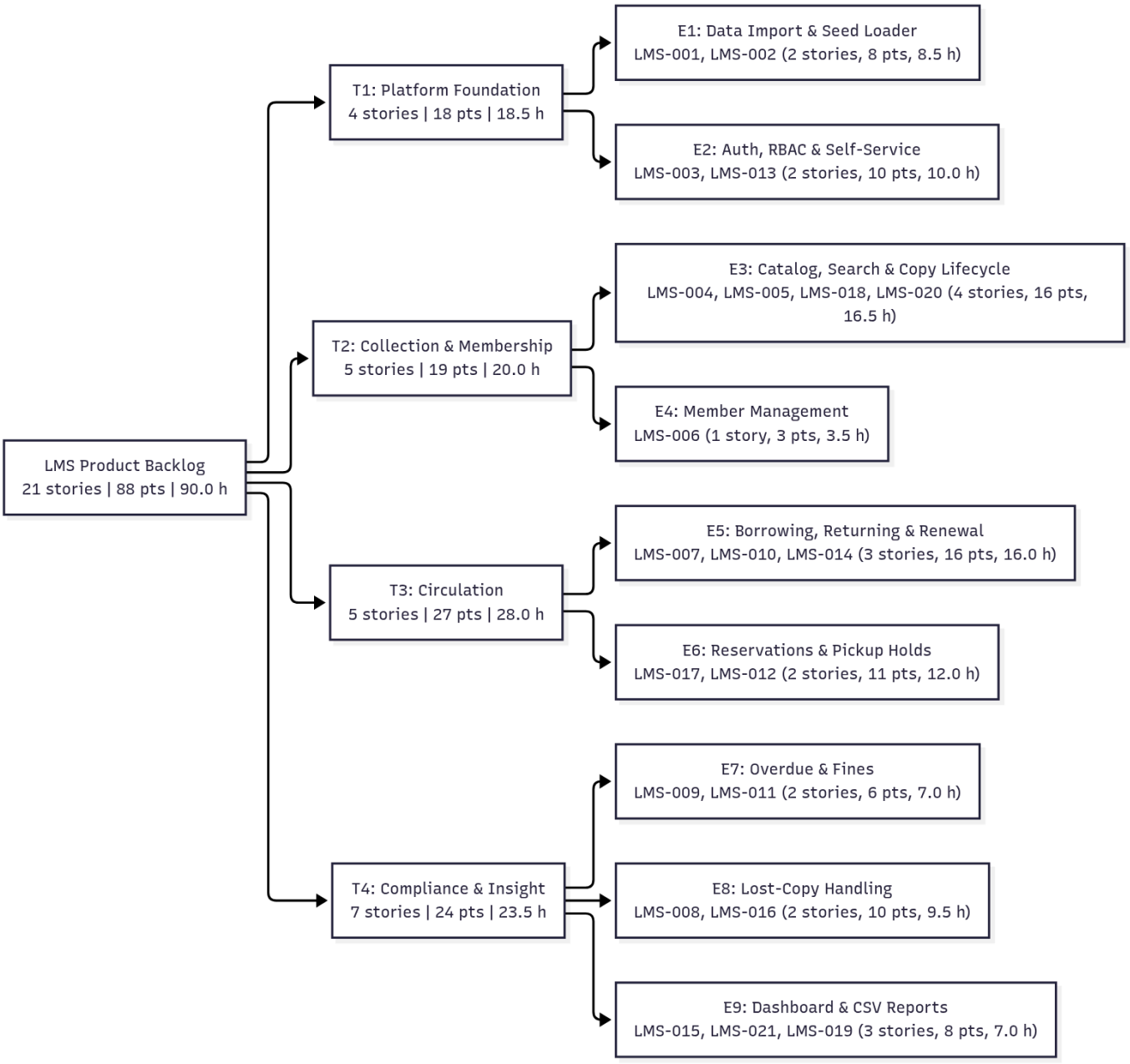
Ubiquitous Term	Seed File (data/seed/)	Natural Key & Core Attributes	Key Domain Invariant
Book	books-openlibrary.json (782)	isbn13 (13-digit ISBN), title, subtitle, authors, publisher, firstPublishYear, category (11), subjects, pageCount, coverId, coverUrl, openLibraryWorkKey, referenceOnly	Owns 1..4 physical BookCopy items (A1). If referenceOnly == true (29 books), copies cannot be loaned or reserved (BR-014).
BookCopy	copies.json (1,706)	barcode (LMS-COPY-1001 ..), isbn13 (FK), status (AVAILABLE, LOANED, RESERVED, MAINTENANCE, LOST)	Governed by BookCopyState; at most one ACTIVE loan or one READY reservation per copy at any moment (BR-017).
Member	members.json (60)	memberId (SYN-STU-*, SYN-STF-*, SYN-ADM-*), firstName, lastName, email (@synthetic.lms.local), memberType, status (ACTIVE, SUSPENDED), department, registeredOn	memberId represents the personal university ID card (BR-016). Quotas resolved via LoanPolicyStrategy (7 / 3 / 3, BR-002).
Librarian	librarians.json (3)	username (syn-lib-0001 ..), displayName, email, role (ROLE_LIBRARIAN)	Zero password fields in JSON; BCrypt demo hashes seeded at startup (BR-018). Settles fines (A5).
Loan	loans.json (164)	loanId (LOAN-0001 ..), memberId, copyBarcode, checkoutDate, dueDate, returnDate, status (ACTIVE, RETURNED, LOST), renewalCount (0..1), renewedOn, lostReportedOn	dueDate = checkoutDate + 30 * (1 + renewalCount). Renewed return triggers 30-day cooldown (BR-006). Lost petition freezes fine (BR-015).
Fine	fines.json (47)	fineId (FINE-0001 ..), loanId (unique FK), memberId, overdueDays, amount (0.25 * overdueDays), currency (TRY), status (UNPAID, PAID, WAIVED), assessedOn, settledOn, settledBy	Uncapped 0.25 TRY/day (BR-008, BR-009). Any UNPAID fine blocks new checkouts and renewals (BR-011).
Reservation	reservations.json (22)	reservationId (RES-0001 ..), isbn13, memberId, queuePosition (1..n), status (WAITING, READY), createdOn, readyCopyBarcode, readyOn, holdUntil	Title-level FIFO queue when 0 copies are AVAILABLE (BR-012); READY hold binds RESERVED copy for 3 days (holdUntil = readyOn + 3 days, BR-013).

Part 2 — Agile Scrum Product Backlog & Iteration Plan

2.1 Themes & Epics (T1 – T4, E1 – E9)

Full specification: docs/02-inception/THEMES_EPICS.md. The product is organized into **4 Themes** and **9 Epics** decomposing into **21 User Stories** (88 Story Points, 173.0 classic hours, 90.0 human hours) and **105 Technical Tasks**.

Theme	Epic	Goal	Stories	Points	Classic h	Human h	Assigned Story IDs (Sprint)
T1 Platform Foundation	—	Data and access foundations every feature relies on	**4**	**18**	**36**	**18.5**	—
↳ T1	**E1 Data import & seed loader**	Load seed v2 reproducibly with dates shifted to startup day (OPEN-04).	2	8	16	8.5	LMS-001 (S1), LMS-002 (S1)
↳ T1	**E2 Auth, RBAC & self-service**	Local login with BCrypt demo passwords (OPEN-05), RBAC, member self-service.	2	10	20	10.0	LMS-003 (S1), LMS-013 (S2)
T2 Collection & Membership	—	Who and what the library holds	**5**	**19**	**38**	**20.0**	—
↳ T2	**E3 Catalog, search & copy lifecycle**	Find books, maintain books/copies, enforce BookCopyState & referenceOnly .	4	16	32	16.5	LMS-004 (S1), LMS-005 (S1), LMS-018 (S3), LMS-020 (S3)
↳ T2	**E4 Member management**	Register, edit, and suspend synthetic members across 3 member types.	1	3	6	3.5	LMS-006 (S2)
T3 Circulation	—	Lending, returning, renewing, and holding books	**5**	**27**	**55**	**28.0**	—
↳ T3	**E5 Borrowing, returning & renewal**	Desk checkout, return, and single renewal under confirmed ADU rules.	3	16	32	16.0	LMS-007 (S2), LMS-010 (S2), LMS-014 (S3)
↳ T3	**E6 Reservations & pickup holds**	Title-level FIFO holds, in-app notification, 3-day pickup window, cancellation/expiry.	2	11	23	12.0	LMS-017 (S3), LMS-012 (S3)
T4 Compliance & Insight	—	Fines, lost copies, and management information	**7**	**24**	**44**	**23.5**	—
↳ T4	**E7 Overdue & fines**	Deterministic overdue days, uncapped 0.25 TRY/day fines, desk payment & waiver.	2	6	13	7.0	LMS-009 (S2), LMS-011 (S3)
↳ T4	**E8 Lost-copy handling**	Design spike (LMS-008) + lost-copy petition, 30-day window, 2x runtime appraisedValue .	2	10	16	9.5	LMS-008 (S1), LMS-016 (S3)
↳ T4	**E9 Dashboard & CSV reports**	Operational KPI cards (015a), overdue/unpaid report (015b), CSV export.	3	8	15	7.0	LMS-015 (S3), LMS-021 (S3), LMS-019 (S3)
Total	**9 Epics**	**Complete Product Backlog**	**21**	**88**	**173**	**90.0**	**11 Must, 7 Should, 3 Could**



2.2 User Stories, Estimation Model & Task Breakdown (`LMS-001` .. `LMS-021`)

Full Given/When/Then acceptance criteria for all 21 stories: `docs/02-inception/USER_STORIES.md`. Full breakdown of all 105 technical tasks: `docs/02-inception/TASKS.md`.

Estimation Methodology (Product Owner Approved)

1. **Fibonacci Story Points (`1, 2, 3, 5, 8`; total `88 pts`):** Relative complexity, rule density, and cross-entity touchpoints (`Must = 48 pts`, `Should = 32 pts`, `Could = 8 pts`). Three closely coupled story pairs from the initial roadmap were combined without inflating points (`Login+RBAC = 5 pts`, `Book CRUD+Copy State = 8 pts`, `Checkout+Cooldown = 8 pts`), and the dashboard story was split into `LMS-015` (`015a` KPI cards, `2 pts`) and `LMS-021` (`015b` overdue/unpaid report table, `3 pts`) so the commitment line lands cleanly inside Sprint 3 net capacity.
2. **Classic Effort Hours (`Classic h`; total `173.0 h`, Must `98.0 h`):** Traditional two-student implementation time without AI coding assistants (`~2.0 h/pt`).
3. **Human Effort Hours (`Human h`; total `90.0 h`, Must `53.0 h`, Should `29.5 h`, Could `7.5 h`):** The actual human time of the two students—writing prompts, steering AI agents, reviewing generated code/docs, executing manual and automated tests, and reviewing Stitch UI screens. Agent runtime is **not** counted. Calculated from `~0.40 x` classic coding effort plus fixed human review/testing overhead, a `4.0 h` timeboxed design spike (`LMS-008`), and PO-approved safety reserves on the two most rule-dense stories (`LMS-005 = 7.5 h`, `LMS-007 = 9.0 h`). **All estimates are explicit planning assumptions to be re-calibrated with Sprint 1 actual hours.**
4. **Agent Sessions (`46` sessions):** Budgeted heavy AI coding/design sessions (`1..4` per story) tracked against weekly AI quota limits (`RISK-H03`).
5. **Task Breakdown (`105` tasks = `90.0` human h in `docs/02-inception/TASKS.md`):** Every story is decomposed into `1.0`–`4.0 h` technical tasks covering Stitch MCP UI prototyping (`NFR-001`, scheduled strictly before Thymeleaf view tasks),

domain/service/repository coding, and JUnit 5 TDD unit/integration tests (NFR-011), with an empty Actual human h column ready for Sprint 1–3 tracking.

Catalog of User Stories (LMS-001 .. LMS-021)

ID	Story Title & User Story Statement (*As a … I want … so that …*)	Key Acceptance Criteria Highlights (Given / When / Then)	Traceability (FR / UC / BR)	Est. (Pts / Classic / Human h / Tasks)
LMS-001	**Domain model and persistence skeleton:** As a developer, I want the 7 JPA entities (Book , BookCopy , Member , Librarian , Loan , Fine , Reservation), 6 enums, and Spring Data repositories mapped to H2 so that all stories build on a tested persistence layer.	Schema starts cleanly on file H2 (./data/db/lms) and in-memory test H2 (jdbc:h2:mem:lms_test); Book (9780002615075) with BookCopy (LMS-COPY-1002), Member (SYN-STU-0001), and Loan (LOAN-0099) round-trip with all associations (A1 .. A8); unique barcode and Fine.loanId constraints reject duplicates.	FR-001 , FR-004 .. FR-006 , FR-011 , FR-015 , FR-019 · BR-010 , BR-017	3 pts / 6 h / 3.5 h (5 tasks)
LMS-002	**Seed loader with date shifting (OPEN-04):** As a librarian/evaluator, I want the app to load data/seed/*.json on startup when H2 is empty and shift all seed dates by delta = today - 2026-10-22 so that active, due-soon, and overdue scenarios stay fresh on any demo date.	Empty H2 loads exact seed v2 counts (782 books with 29 referenceOnly , 1,706 copies, 60 members, 3 librarians, 164 loans, 47 fines, 22 reservations); non-empty H2 skips reload (idempotent); date shift preserves overdueDays (FINE-0022 on LOAN-0107) and holdUntil - readyOn == 3 days (RES-0009).	FR-019 , DR-001 .. DR-007 · BR-017 , BR-018	5 pts / 10 h / 5.0 h (5 tasks)
LMS-003	**Login/logout with BCrypt demo passwords (OPEN-05) and RBAC:** As a librarian or member, I want to log in with a startup-seeded BCrypt password and see only endpoints permitted for my role so that desk operations are protected.	data/seed/*.json contains zero password fields; startup seeds BCrypt hashes for syn-lib-0001 .. 0003 (ROLE_LIBRARIAN) and SYN-STU-* / SYN-STF-* / SYN-ADM-* (ROLE_MEMBER); ROLE_MEMBER gets HTTP 403 on librarian CRUD/desk URLs; invalid credentials show safe error; logout invalidates session + CSRF.	FR-016 , FR-017 , NFR-007 , NFR-008 · UC-02 · BR-001 , BR-018	5 pts / 10 h / 5.0 h (6 tasks)
LMS-004	**Catalog search and book detail:** As a member or librarian, I want to search books by partial title, author, or exact isbn13 , filter by category, availability, or referenceOnly , and inspect copy states on the detail page.	Query "Yaşar Kemal" or "9780002615075" returns *İnce Memed* with available/total copy counts; referenceOnly == true (9780824722876) shows REFERENCE ONLY — NOT FOR LOAN badge and disables borrow/reserve; detail page lists all copies (LMS-COPY-1002 .. 1004) and active hold queue count.	FR-002 , FR-003 · UC-01 · BR-001 , BR-014 , BR-017	3 pts / 6 h / 3.5 h (5 tasks)
LMS-005	**Book CRUD, copy registration and copy lifecycle:** As a librarian, I want to create/edit/archive Book records, register LMS-COPY-XXXX barcodes, and enforce BookCopyState transitions so that inventory stays accurate.	Invalid ISBN-13 checksum is rejected; valid book creation persists all fields including referenceOnly ; new copy accepts AVAILABLE / MAINTENANCE and rejects duplicate barcode; BookCopyState allows AVAILABLE <-> MAINTENANCE and blocks LOANED -> MAINTENANCE or checkout of LOANED / MAINTENANCE / LOST .	FR-001 , FR-004 , FR-005 · UC-03 · BR-014 , BR-015 , BR-017	8 pts / 16 h / 7.5 h (7 tasks)
LMS-008	**Design spike — compensation tracking & reservation terminal states:** As an architect, I want a timeboxed spike resolving HW01 §3.7 open design items before Sprint 3 coding starts so that LMS-016 and LMS-017 require zero schema rework.	Produces ADR in docs/03-design/ specifying (a) how 2.00 * appraisedValue lost-copy compensation is persisted and blocks borrowing under BR-011 , and (b) whether fulfilled/cancelled/expired reservations use terminal enum values or deletion; updates JPA skeleton with green tests.	FR-011 , FR-023 , FR-027 · UC-05 , UC-08 , UC-09 · BR-011 .. BR-013 , BR-015	2 pts / 4 h / 4.0 h (3 tasks)
LMS-006	**Member registration, profile update and suspension:** As a librarian, I want to register synthetic STUDENT , ACADEMIC_STAFF , and ADMINISTRATIVE_STAFF members, edit profiles, and toggle ACTIVE / SUSPENDED status.	Validates SYN-STU-* / SYN-STF-* / SYN-ADM-* ID and @synthetic.lms.local email (NFR-009); displays confirmed quota (7 / 3 / 3); toggling status to SUSPENDED immediately blocks new checkouts, renewals, and reservations (BR-011).	FR-006 , FR-007 · UC-04 · BR-001 , BR-002 , BR-011 , BR-018	3 pts / 6 h / 3.5 h (5 tasks)
LMS-007	**Checkout with eligibility, cooldown and referenceOnly enforcement:** As a librarian, I want to check out an eligible copy to an in-person member (memberId + copyBarcode) with full validation of ADU rules in one @Transactional step.	LoanPolicyStrategy enforces max 7 (ACADEMIC_STAFF) vs 3 (ADMINISTRATIVE_STAFF / STUDENT) and dueDate = checkoutDate + 30 days ; blocks SUSPENDED , overdue active loan (LOAN-0099), UNPAID fine (FINE-0022), referenceOnly == true (9780824722876), non-AVAILABLE copy, or active 30-day post-renewal cooldown (BR-006).	FR-009 .. FR-011 , FR-013 · UC-05 · BR-002 , BR-003 , BR-006 , BR-011 , BR-013 , BR-014 , BR-016 , BR-017	8 pts / 18 h / 9.0 h (7 tasks)

ID	Story Title & User Story Statement (*As a … I want … so that …*)	Key Acceptance Criteria Highlights (Given / When / Then)	Traceability (FR / UC / BR)	Est. (Pts / Classic / Human h / Tasks)
LMS-009	**Overdue detection and uncapped fine calculation:** As a librarian, I want deterministic overdue days (effectiveEndDate > dueDate, zero grace) and uncapped 0.25 TRY/day fines (FineCalculationStrategy) so penalties match ADU rules.	effectiveEndDate == dueDate yields 0 overdue days and no fine; 1 day late yields 0.25 TRY UNPAID; 1,200 days late yields 300.00 TRY (no 250 TRY cap, BR-009); lostReportedOn freezes overdueDays as of petition date (LOAN-0103, BR-015).	FR-014, FR-015 · UC-06, UC-09, UC-11 · BR-007..BR-010, BR-015	3 pts / 6 h / 3.5 h (4 tasks)
LMS-010	**Return of a copy (AVAILABLE-only in Sprint 2):** As a librarian, I want to scan a returned copyBarcode to close its ACTIVE loan, finalize any overdue fine, record post-renewal cooldown anchor, and transition the copy to AVAILABLE.	Atomic @Transactional return sets returnDate = currentDate, status = RETURNED, transitions copy -> AVAILABLE (hold promotion to RESERVED is added in Sprint 3 by LMS-017), persists/updates UNPAID fine if returnDate > dueDate, and anchors 30-day cooldown if renewalCount >= 1.	FR-012..FR-015 · UC-06 · BR-006..BR-010, BR-017	3 pts / 6 h / 3.5 h (5 tasks)
LMS-013	**Member self-service portal & librarian member inspection:** As a member (or librarian inspecting a member), I want to view active loans, due dates, historical loans, 30-day cooldown timers, and fines (UNPAID / PAID / WAIVED).	Authenticated member (SYN-STU-0001) sees own active/historical loans, cooldown end dates, and fines (LOAN-0099) and cannot access another member's profile (403); Librarian (syn-lib-0001) inspects any member with quota (7 / 3 / 3) and block reason banner. *(Holds/notifications added in LMS-017).*	FR-008, FR-018 · UC-04, UC-10 · BR-001, BR-002, BR-006, BR-010, BR-011	5 pts / 10 h / 5.0 h (6 tasks)
LMS-017	**Reservation, FIFO queue, hold promotion & in-app notification:** As a member or librarian, I want to place a FIFO hold when 0 copies are AVAILABLE, and have a returned copy automatically held as RESERVED (READY) for 3 days with an in-app alert.	Rejects hold if >= 1 copy is AVAILABLE, referenceOnly == true, member blocked (BR-011), or already borrowing/queuing isbn13; assigns FIFO queuePosition = 1..n; on return (UC-06 Observer), promotes queuePosition == 1 to READY (readyCopyBarcode, holdUntil = readyOn + 3 days), sets copy -> RESERVED, creates in-app notification, and allows checkout of RESERVED copy only to queuePosition == 1 member.	FR-010..FR-012, FR-018, FR-020..FR-022 · UC-05, UC-06, UC-08, UC-10 · BR-011..BR-014, BR-017	8 pts / 16 h / 8.5 h (7 tasks)
LMS-011	**Fine payment (PAID) and waiver (WAIVED):** As a librarian, I want to record desk payment or administrative waiver of an UNPAID fine on a resolved loan so that the member's financial block is cleared with auditability.	Marks Fine.status = PAID or WAIVED with settledOn = currentDate and settledBy = librarian.username (syn-lib-0001); rejects settlement while loan is still ACTIVE; clears BR-011 block once all overdue loans and UNPAID balances reach zero.	FR-025, FR-026 · UC-09 · BR-010, BR-011	3 pts / 7 h / 3.5 h (5 tasks)
LMS-012	**Reservation cancellation and 3-day hold expiry:** As a member or librarian, I want to cancel a reservation and have unclaimed READY holds expire after 3 days (currentDate > holdUntil) so RESERVED copies are never locked forever.	Cancelling WAITING decrements trailing queuePositions; cancelling or expiring a READY hold promotes the next WAITING reservation (rebinding readyCopyBarcode and holdUntil = currentDate + 3 days) or returns copy.status to AVAILABLE if queue is empty.	FR-023 · UC-08 · BR-012, BR-013, BR-017	3 pts / 7 h / 3.5 h (5 tasks)
LMS-015	**Operational dashboard KPI summary cards (015a):** As a librarian, I want a dashboard with real-time KPI cards (titles, copies by state, active loans, due-soon 0..3 days, overdue loans, total UNPAID fine TRY) for instant health visibility.	Seeded dataset renders 782 titles, 1,706 copies broken down across all 5 states, active loan count, due-soon count (dueDate in [currentDate, currentDate + 3 days]), overdue active count, and exact sum of UNPAID fines in TRY.	FR-028 · UC-11 · BR-001, BR-007, BR-008	2 pts / 4 h / 2.0 h (4 tasks)
LMS-014	**Loan renewal with reserved guard (Stretch):** As a member or librarian, I want to renew an eligible ACTIVE loan once (+30 days) when not overdue, not already renewed, and not reserved by another member.	renewalCount: 0 -> 1, renewedOn = currentDate, dueDate = checkoutDate + 60 days; rejects if renewalCount >= 1, currentDate > dueDate, any WAITING / READY reservation exists on isbn13, or borrower is blocked (BR-011).	FR-024 · UC-07 · BR-003..BR-006, BR-011	5 pts / 8 h / 3.5 h (5 tasks)
LMS-016	**Lost-copy process with runtime appraisedValue (Stretch):** As a librarian, I want to record a lost-book petition (lostReportedOn), freeze daily overdue fines, and after 30 days enter appraisedValue to bill 2.00 * appraisedValue.	Recording lostReportedOn freezes overdueDays; replacement within 30 days closes loan as RETURNED; after 30 days, Librarian enters appraisedValue (e.g., 450.00 TRY), system charges 900.00 TRY (2x) + frozen fine, sets copy -> LOST and loan -> LOST (returnDate = null).	FR-005, FR-014, FR-027 · UC-09 · BR-007, BR-011, BR-015, BR-017	8 pts / 12 h / 5.5 h (6 tasks)

ID	Story Title & User Story Statement (*As a … I want … so that …*)	Key Acceptance Criteria Highlights (`Given / When / Then`)	Traceability (`FR / UC / BR`)	Est. (`Pts / Classic / Human h / Tasks`)
`LMS-021`	**Actionable overdue loans & unpaid fines report (`015b`, *Stretch*):** As a librarian, I want a prioritized table of overdue active loans and `UNPAID` fines with direct links to return/settle so I can act on delinquencies.	Displays `loanId`, `memberId`, member name, `copyBarcode`, `title`, `dueDate`, `overdueDays` (descending), accrued `amount` (`0.25 TRY/day`), and quick action links to member profile, return desk, and fine settlement.	`FR-014`, `FR-015`, `FR-029` · `UC-11` · `BR-001`, `BR-007`, `BR-008`, `BR-011`	`3 pts` / `6 h` / `3.0 h` (`3` tasks)
`LMS-018`	**Public guest catalog portal (`C1`, *Stretch*):** As an unauthenticated guest (`ROLE_GUEST`), I want to browse/search the catalog and view copy availability in read-only mode without seeing any member or loan data.	Anonymous `GET /catalog` and `/catalog/{isbn13}` succeed (`200 OK`) showing availability and `referenceOnly` badges; zero member/loan PII is rendered; reserve/borrow buttons redirect to `/login`.	`FR-002`, `FR-003`, `FR-030` · `UC-01` · `BR-001`, `BR-014`	`2 pts` / `4 h` / `2.0 h` (`4` tasks)
`LMS-019`	**CSV export of overdue & inventory reports (`C3`, *Stretch*):** As a librarian, I want to download the overdue loans report and physical copy inventory as UTF-8 CSV files for offline auditing.	`ROLE_LIBRARIAN` downloads valid UTF-8 CSV (`text/csv`) with escaped fields and Turkish diacritics (`ğ, ş, ı, ö, ü, ç`) intact; non-librarians receive HTTP `403`.	`FR-032` · `UC-11` · `BR-001`	`3 pts` / `5 h` / `2.0 h` (`4` tasks)
`LMS-020`	**Open Library ISBN lookup helper (`C2`, *Stretch*):** As a librarian adding a book, I want an optional "Fetch by ISBN" helper (`OpenLibraryCatalogAdapter`) that pre-fills metadata and degrades gracefully when offline.	Online lookup populates title, authors, publisher, year, and cover URL; offline or timeout (`<= 3 s`) shows a non-blocking banner (*"Offline mode: please enter book metadata manually"*) keeping manual form entry 100% functional (`NFR-004`).	`FR-001`, `FR-031` · `UC-03` · `BR-001`	`3 pts` / `6 h` / `3.5 h` (`4` tasks)

2.3 Ordered Product Backlog & End-to-End Traceability

Authoritative source: `docs/02-inception/PRODUCT_BACKLOG.md`, validated by `python3 tools/agile/validate_backlog.py`.

Ordered Product Backlog (`LMS-001` .. `LMS-021`)

Order	ID	Epic	Story Title	MoSCoW	Pts	Classic h	Human h	Agent Sess.	Sprint	Depends On	Status
1	`LMS-001`	E1	Domain model and persistence skeleton	Must	3	6	3.5	2	S1	—	Planned (S1)
2	`LMS-002`	E1	Seed loader with date shifting	Must	5	10	5.0	3	S1	`LMS-001`	Planned (S1)
3	`LMS-003`	E2	Login/logout with BCrypt demo passwords and RBAC	Must	5	10	5.0	3	S1	`LMS-001`, `LMS-002`	Planned (S1)
4	`LMS-004`	E3	Catalog search and book detail	Must	3	6	3.5	2	S1	`LMS-002`, `LMS-003`	Planned (S1)
5	`LMS-005`	E3	Book CRUD, copy registration and copy lifecycle	Must	8	16	7.5	3	S1	`LMS-003`, `LMS-004`	Planned (S1)
6	`LMS-008`	E8	Design spike: compensation tracking and reservation terminal states	Must	2	4	4.0	1	S1	`LMS-001`	Planned (S1)
7	`LMS-006`	E4	Member registration, profile update and suspension	Must	3	6	3.5	2	S2	`LMS-003`	Planned (S2)
8	`LMS-007`	E5	Checkout with eligibility, cooldown and referenceOnly enforcement	Must	8	18	9.0	4	S2	`LMS-005`, `LMS-006`	Planned (S2)
9	`LMS-009`	E7	Overdue detection and uncapped fine calculation	Must	3	6	3.5	2	S2	`LMS-001`	Planned (S2)
10	`LMS-010`	E5	Return of a copy (copy to AVAILABLE only)	Must	3	6	3.5	2	S2	`LMS-007`, `LMS-009`	Planned (S2)
11	`LMS-013`	E2	Member self-service portal and librarian member inspection	Must	5	10	5.0	3	S2	`LMS-007`, `LMS-009`, `LMS-010`	Planned (S2)
12	`LMS-017`	E6	Reservation, FIFO queue, hold promotion and in-app notification	Should	8	16	8.5	4	S3	`LMS-010`, `LMS-008`	Planned (S3: **commit**)

Order	ID	Epic	Story Title	MoSCoW	Pts	Classic h	Human h	Agent Sess.	Sprint	Depends On	Status
13	LMS-011	E7	Fine payment and waiver	Should	3	7	3.5	2	S3	LMS-009, LMS-010	Planned (S3: **commit**)
14	LMS-012	E6	Reservation cancellation and hold expiry	Should	3	7	3.5	2	S3	LMS-017	Planned (S3: **commit**)
15	LMS-015	E9	Operational dashboard KPI summary cards (015a)	Should	2	4	2.0	1	S3	LMS-009, LMS-010	Planned (S3: **commit**)
16	LMS-014	E5	Loan renewal with reserved guard	Should	5	8	3.5	2	S3	LMS-007	Planned (S3: *stretch*)
17	LMS-016	E8	Lost-copy process with runtime appraised value	Should	8	12	5.5	3	S3	LMS-010, LMS-011, LMS-008	Planned (S3: *stretch*)
18	LMS-021	E9	Actionable overdue loans and unpaid fines report (015b)	Should	3	6	3.0	1	S3	LMS-009, LMS-010, LMS-015	Planned (S3: *stretch*)
19	LMS-018	E3	Public guest catalog portal	Could	2	4	2.0	1	S3	LMS-004	Planned (S3: *stretch*)
20	LMS-019	E9	CSV export of overdue and inventory reports	Could	3	5	2.0	1	S3	LMS-021	Planned (S3: *stretch*)
21	LMS-020	E3	Open Library ISBN lookup helper	Could	3	6	3.5	2	S3	LMS-005	Planned (S3: *stretch*)
Total	**21**	**E1–E9**	**11 Must (48 pts / 53.0 h), 7 Should (32 pts / 29.5 h), 3 Could (8 pts / 7.5 h)**	—	**88**	**173**	**90.0**	**46**	**S1–S3**	**0 forward deps**	**105 tasks (90.0 h)**

End-to-End Traceability Matrix (FR-001 .. FR-032 -> UC -> BR -> Story -> Sprint)

Every Must Have functional requirement (FR-001 .. FR-019) is assigned to a story in Sprint 1 or Sprint 2 (Must FR not covered in S1+S2: none).

FR-* ID	MoSCoW Priority	Realizing Use Case(s)	Enforced Business Rule(s)	Implementing Story	Target Sprint
FR-001, FR-004, FR-005	Must Have (M1)	UC-03, UC-05, UC-06, UC-09	BR-014, BR-015, BR-017	LMS-005	**S1**
FR-002, FR-003	Must Have (M1)	UC-01	BR-001, BR-014, BR-017	LMS-004	**S1**
FR-006, FR-007	Must Have (M2)	UC-04	BR-001, BR-002, BR-011, BR-018	LMS-006	**S2**
FR-008, FR-018	Must Have (M2, M5)	UC-04, UC-10	BR-001, BR-002, BR-006, BR-010, BR-011, BR-013	LMS-013	**S2**
FR-009, FR-010, FR-011, FR-013	Must Have (M3)	UC-05, UC-06	BR-002, BR-003, BR-006, BR-011, BR-013, BR-014, BR-016, BR-017	LMS-007	**S2**
FR-012	Must Have (M3)	UC-06	BR-007, BR-008, BR-013, BR-017	LMS-010	**S2**
FR-014, FR-015	Must Have (M4)	UC-06, UC-09, UC-11	BR-007, BR-008, BR-009, BR-010, BR-015	LMS-009	**S2**
FR-016, FR-017	Must Have (M5)	UC-02	BR-001, BR-018	LMS-003	**S1**
FR-019	Must Have (M6)	System Startup	BR-017, BR-018	LMS-002 (on LMS-001)	**S1**
FR-020, FR-021, FR-022	Should Have (S1)	UC-06, UC-08, UC-10	BR-011, BR-012, BR-013, BR-014	LMS-017	**S3 (Commit)**
FR-023	Should Have (S1)	UC-08	BR-012, BR-013	LMS-012	**S3 (Commit)**

FR-* ID	MoSCoW Priority	Realizing Use Case(s)	Enforced Business Rule(s)	Implementing Story	Target Sprint
FR-024	Should Have (S2)	UC-07	BR-003, BR-004, BR-005, BR-011	LMS-014	**S3 (Stretch)**
FR-025, FR-026	Should Have (S3)	UC-09	BR-010, BR-011	LMS-011	**S3 (Commit)**
FR-027	Should / Could (S3)	UC-09	BR-005, BR-015, BR-017	LMS-016 (on LMS-008)	**S3 (Stretch)**
FR-028	Should Have (S4)	UC-11	BR-001, BR-007, BR-008	LMS-015 (015a)	**S3 (Commit)**
FR-029	Should Have (S4)	UC-11	BR-007, BR-008, BR-011	LMS-021 (015b)	**S3 (Stretch)**
FR-030	Could Have (C1)	UC-01	BR-001, BR-014	LMS-018	**S3 (Stretch)**
FR-031	Could Have (C2)	UC-03	BR-001	LMS-020	**S3 (Stretch)**
FR-032	Could Have (C3)	UC-11	BR-001	LMS-019	**S3 (Stretch)**

2.4 Three-Sprint Iteration Plan, Capacity Model & Commitment Line

Authoritative source: `docs/02-inception/SPRINT_PLAN.md`.

Sprint Schedule & Capacity Model

- **Capacity Unit (`Human h`):** Two students' human effort (`8 h/person/week` in normal weeks, reduced to `3 h/person/week` during midterm weeks `Nov 17 – Nov 28`, prorated per calendar day).
- **Fixed Deductions per Sprint:** Scrum ceremonies = `4.0 h` (`2 people x 2 h`); Homework engineering/design documentation = `8.0 h` in Sprint 1 (`HW03`: Use Case diagram `1.5 h`, System Sequence Diagrams `2.0 h`, Class Diagram `2.0 h`, TDD/Git/Maven plan `1.0 h`, Stitch UI review `1.5 h`), `4.0 h` in Sprint 2 (`HW04`), and `7.0 h` in Sprint 3 (`HW05`: user guide, test report, final retrospective); Contingency buffer = **`10%` of gross capacity** (never reduced).

Sprint	Calendar Window	Course Deliverable	Gross (h)	Ceremonies (h)	HW Docs (h)	10% Buffer (h)	Net Story Capacity (h)	Planned Stories (h)	Must (h)	Should (h)	Could (h)	Planned - Net
Sprint 1	Nov 11 – Dec 08 (28 d, incl. midterms Nov 17–28)	HW03 (Elaboration + S1)	46.9	4.0	8.0	4.7	**30.2**	28.5 (26 pts)	28.5	0.0	0.0	-1.7 h (fits)
Sprint 2	Dec 09 – Dec 22 (14 d)	HW04 (Construction + S2)	32.0	4.0	4.0	3.2	**20.8**	24.5 (22 pts)	24.5	0.0	0.0	+3.7 h
Sprint 3	Dec 23 – Jan 05 (14 d)	HW05 (Transition + MVP)	32.0	4.0	7.0	3.2	**17.8**	17.5 committed (+19.5 stretch)	0.0	29.5	7.5	-0.3 h committed (+19.2 h w/ stretch)
Total	**Nov 11 – Jan 05**	**HW03 – HW05**	**110.9**	**12.0**	**19.0**	**11.1**	**68.8**	**70.5 committed (90.0 all)**	**53.0**	**29.5**	**7.5**	**S1+S2 Must gap: +2.0 h**

Must Gap Analysis (`+2.0 h, 4.0%` of S1+S2 Net Capacity — Accepted Risk `RISK-H01`)

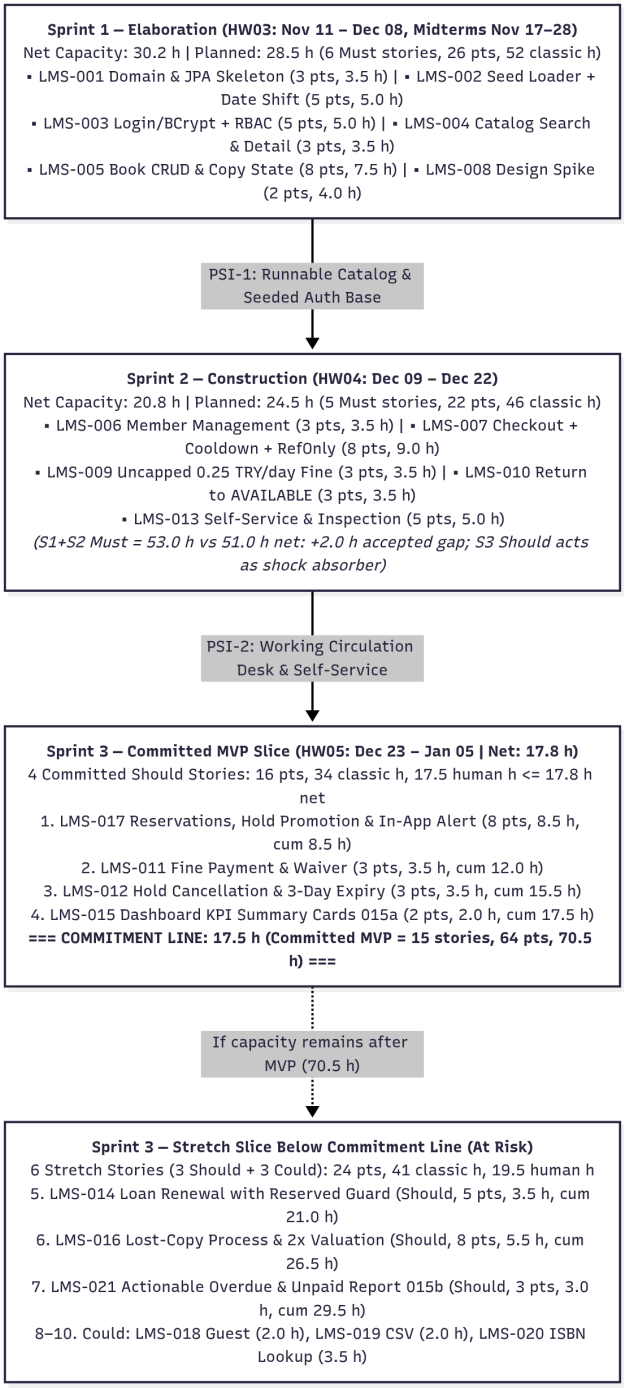
All `11` Must stories require **`53.0` human hours** (`28.5 h` in S1 + `24.5 h` in S2) against a combined S1+S2 net story capacity of **`51.0` hours** (`30.2 + 20.8 h`), leaving a **known gap of `+2.0 h` (`4.0%` of net)**. Rather than artificially lowering the estimates of complex stories (`LMS-005 7.5 h`, `LMS-007 9.0 h`) or cutting the 10% buffer, the Product Owner approved three governance rules:

1. **Buffer Preservation:** The 10% buffer (`4.7 h` in S1, `3.2 h` in S2, `3.2 h` in S3) is **not** reduced.
2. **Sprint 3 Should Stories as Shock Absorber:** If Must stories overrun in Sprint 2, the overrun is absorbed by deferring Sprint 3 Should stories from the bottom of the commitment line upward.
3. **Quantitative Re-Plan Trigger:** If at the end of Sprint 1 the actual ratio `actual_human_h / classic_h` exceeds **`0.45`**, or if the remaining Sprint 2 Must estimate exceeds Sprint 2 net capacity (`20.8 h`) by more than **`10%`**, the Sprint 2 plan is re-baselined and any scope decision is brought to the Product Owner.

Sprint 3 Value Ordering & Commitment Line (`17.5 h <= 17.8 h` Net Capacity)

Because total Sprint 3 Should (`29.5 h`) + Could (`7.5 h`) = `37.0 h` exceeds Sprint 3 net capacity (`17.8 h`), the Product Owner ordered the Sprint 3 stories by domain value and state-machine consistency and drew an explicit **commitment line at `17.5 h`**:

S3 Rank	Story ID & Title	MoSCoW	Pts	Classic h	Human h	Cumulative Human h	Commitment Status & Domain Rationale
1	`LMS-017` Reservation, FIFO queue, hold promotion & in-app notification	Should (`S1`)	8	16	`8.5`	`8.5 h`	**COMMIT** — Core FIFO hold queue, `RESERVED` copy promotion on return, and 3-day pickup notification (`FR-020` .. `FR-022` , `BR-012` , `BR-013`).
2	`LMS-011` Fine payment and waiver	Should (`S3`)	3	7	`3.5`	`12.0 h`	**COMMIT** — Desk `PAID` / `WAIVED` settlement (`FR-025` , `FR-026` , `BR-010`). Required so members with fines accrued in S2 can clear their `BR-011` borrowing block.
3	`LMS-012` Reservation cancellation and hold expiry	Should (`S1`)	3	7	`3.5`	`15.5 h`	**COMMIT** — Cancellation & 3-day hold expiry reallocation (`FR-023`). Required for reservation state-machine completeness so an unclaimed `READY` hold never locks a `RESERVED` copy forever.
4	`LMS-015` Operational dashboard KPI summary cards (`015a`)	Should (`S4`)	2	4	`2.0`	**`17.5 h <= 17.8 h`**	**COMMIT (Commitment Line)** — Delivers real-time librarian KPI cards (`FR-028` : titles, copies by state, active/due-soon/overdue loans, unpaid fine total).
5	`LMS-014` Loan renewal with reserved guard	Should (`S2`)	5	8	`3.5`	`21.0 h`	*STRETCH (below line)* — Single `+30` -day self-service/desk renewal (`FR-024` , `BR-004` , `BR-005`).
6	`LMS-016` Lost-copy process with runtime `appraisedValue`	Should (`S3`)	8	12	`5.5`	`26.5 h`	*STRETCH (below line)* — Lost-book petition fine freeze, 30-day window, and `2.00 * appraisedValue` runtime charge (`FR-027` , `BR-015`).
7	`LMS-021` Actionable overdue loans & unpaid fines report (`015b`)	Should (`S4`)	3	6	`3.0`	`29.5 h`	*STRETCH (below line)* — Detailed overdue/unpaid report table with quick desk links (`FR-029`).
8–10	`LMS-018` Guest portal (`2.0 h`), `LMS-019` CSV export (`2.0 h`), `LMS-020` ISBN lookup (`3.5 h`)	Could (`C1` – `C3`)	8	15	`7.5`	`37.0 h`	*STRETCH (below line)* — Optional enhancements (`FR-030` .. `FR-032`) attempted only if all committed and Should stretch items finish early.



Cross-Sprint Dependency Fixes, DoR, DoD & Risk Register

- **Dependency Hygiene (Verified by `validate_backlog.py` -> `OVERALL: PASS`):** No story depends on a story in a later sprint. Specifically: (1) `LMS-010` (S2 Return) transitions returned copies to `AVAILABLE` only, while `RESERVED` hold promotion is added in S3 by `LMS-017`; (2) `LMS-013` (S2 Self-Service) displays active/historical loans, cooldowns, and fines only, while holds and in-app notifications are added in S3 by `LMS-017`; (3) `LMS-014` (S3 Renewal) checks active reservations via a repository query; and (4) `LMS-008` (S1 Spike) resolves compensation and reservation terminal-state design before S3.
- **Definition of Ready (DoR):** Standard user story format + testable Given/When/Then criteria with `FR` / `UC` / `BR` tags; Fibonacci points, classic/human hours, and agent sessions estimated; zero forward dependencies; Stitch MCP UI task scheduled before any Thymeleaf task; required seed test data identified in `data/seed/*.json`.
- **Definition of Done (DoD):** `./mvnw -q test` (or `tools/mvnw-vm.sh -q test` on the VM) and `python3 -m unittest discover -s tools/data/tests` pass 100% offline; mandatory CSE305 header on all first-party files; `docs/process/ai-usage-log.md` updated with exact model selector name and `"Pending human review"` until explicitly confirmed by a team member; cross-model review completed; UI matches Stitch prototype; merged to `develop` with sprint review/retrospective notes.

Risk ID	Risk Description	Impact	Mitigation & Trigger
`RISK-H01`	**Must gap of `+2.0 h` in S1+S2** (`53.0 h` Must vs `51.0 h` net)	Sprint 2 schedule pressure	Accepted by PO; 10% buffer kept intact; S3 Should stories absorb overrun; re-plan if S1 `human/classic` > `0.45` or S2 Must exceeds net by `> 10%`.

Risk ID	Risk Description	Impact	Mitigation & Trigger
RISK-H02	**Sprint 3 Stretch stories (19.5 h) exceed S3 net (17.8 h)**	Stretch items (LMS-014 , 016 , 021 , 018 .. 020) not reached	Explicit commitment line at 17.5 h <= 17.8 h ; MVP (70.5 h) requires only the 4 committed Should stories above the line.
RISK-H03	**Weekly AI quota exhaustion** across 46 budgeted agent sessions	Interruption mid-story	Commit & push after every file/step; keep docs/process/*_PROGRESS.md updated; budget agent sessions per week.
RISK-H04	**Human-hour conversion ratio (~0.40x) is an assumption**	Velocity variance	Track actual human hours in TASKS.md during Sprint 1 and re-calibrate Sprint 2–3 estimates at Sprint 1 Review.
RISK-H05 .. H08	**Midterm capacity drop (Nov 17–28), VM proxy Maven wrapper, HW01 open design items, dual ADUZEM upload**	Schedule/build/submission issues	Midterms prorated (3 h/wk); tools/mvnw-vm.sh isolated from standard ./mvnw ; LMS-008 spike in S1; mandatory dual-upload check.

2.5 Potentially Shippable Increments (PSI) & MVP Definition

Authoritative source: docs/02-inception/PSI_AND_MVP.md .

Increment	Included Stories	Committed Effort	End-to-End Demonstrable Capability (PSI / MVP)
Sprint 1 PSI (HW03)	LMS-001 .. LMS-005 , LMS-008 (6 Must stories)	26 pts / 52 classic h / **28.5 human h**	Application boots offline, loads seed v2 (782 books, 1,706 copies) with shifted dates (OPEN-04), authenticates Librarians and Members via BCrypt demo passwords (OPEN-05) and RBAC, supports catalog search/filtering and referenceOnly badges, enforces BookCopyState on Book/Copy CRUD, and documents the LMS-008 ADR.
Sprint 2 PSI (HW04)	+ LMS-006 , LMS-007 , LMS-009 , LMS-010 , LMS-013 (5 Must stories)	22 pts / 46 classic h / **24.5 human h** *(Cumulative Must: 48 pts / 53.0 h)*	Complete circulation desk without holds: Librarian registers/suspends members, executes atomic checkouts (7 / 3 / 3 quotas, 30 -day period, referenceOnly & block checks, 30-day post-renewal cooldown) and returns (AVAILABLE), computes uncapped 0.25 TRY/day fines, and Members inspect active/historical loans, cooldowns, and fines.
Sprint 3 PSI — Committed MVP (HW05)	+ LMS-017 , LMS-011 , LMS-012 , LMS-015 (4 Committed Should stories above the 17.5 h line)	16 pts / 34 classic h / **17.5 human h** ***(Total Committed MVP: 15 stories, 64 pts , 132 classic h, 70.5 human h)***	Adds title-level FIFO reservations, automatic RESERVED hold promotion on return with 3-day pickup window and in-app notification (LMS-017), desk fine PAID / WAIVED settlement (LMS-011), reservation cancellation and 3-day hold expiry reallocation (LMS-012), and operational dashboard KPI summary cards (LMS-015).
Sprint 3 Stretch Slice *(Below Line, At Risk)*	LMS-014 , LMS-016 , LMS-021 , LMS-018 .. LMS-020 (3 Should + 3 Could stories)	24 pts / 41 classic h / **19.5 human h** *(Full backlog if all stretch completed: 21 stories, 88 pts , 90.0 h)*	Self-service/desk renewal (LMS-014), lost-copy petition & 2x runtime appraisedValue assessment (LMS-016), actionable overdue/unpaid report table (LMS-021), public guest catalog (LMS-018), CSV export (LMS-019), and live Open Library ISBN lookup helper (LMS-020).

Revision History

Version	Date	Summary
1.0	2026-11-07	First consolidated HW02 submission document: Part 1 consolidates the UP Inception Overview (VISION_AND_CONTEXT.md , PERSONAS.md , REQUIREMENTS.md , BUSINESS_RULES.md , USE_CASES.md , DOMAIN_GLOSSARY.md) with embedded System Context and Use Case diagrams; Part 2 presents the Product Owner-approved Agile Scrum Product Backlog (21 stories LMS-001 .. LMS-021 , 88 story points, 173.0 classic hours, 90.0 human hours, 105 tasks), 3-sprint capacity model, accepted +2.0 h Must gap governance (RISK-H01), Sprint 3 commitment line (17.5 h <= 17.8 h net), Themes/Epics and Sprint Roadmap diagrams, dependency proof (OVERALL: PASS), and PSI/MVP definition (64 pts / 70.5 h committed).